RSM317:
Text Mining and
Natural Language Processing

This course will introduce students to a **diverse collection of text mining techniques** and **natural language processing using machine learning**. These techniques are often aimed at identifying and quantifying various structures in the text data to answer business problems and provide managerial insights. Model validation and effective communication of model-based results will be stressed. The course will employ a "**white-box**" methodology, which emphasizes an understanding of the algorithmic and statistical model structures and how they apply to text analysis. This is a **hands-on** course focused on **developing practical skills** across a broad range of the most common text mining and natural language processing techniques. Following leading industry standards, the course will use **Python** to apply a number of different algorithms to real-world big text data. Because today's generative-AI systems are themselves built on the methods studied here, the course also treats the responsible, well-understood use of such tools—verifying and taking ownership of their output—as a core professional skill.

Administrative Issues

- Getting to know me:
  - Gerhard Trippen
    - » E-mail: Gerhard.Trippen@rotman.utoronto.ca
    - » Please start subject with **RSM317**
    - » Office: RT 406
    - » Wednesdays, 15:00-17:00 (RT 406), and other times by appointment
- Getting to know you:
  - Please bring a "name tent"
  - Please submit a one-page summary of yourself, including your name, student number, email address, your major, a picture of yourself, your interests, and your expectations of the course, etc. Due next class.

Course Materials

- Required materials:
  - Syllabus
  - Class notes (posted on course website: q.utoronto.ca)
- Recommended materials:
  - DataCamp (datacamp.com)
    - » Optional practice assignments in Quercus — not graded:
      - Introduction to Python; Basic Concepts of String Manipulation; Formatting Strings; Regular Expressions for Pattern Matching
  - ANACONDA
    - » Install it to work on the group assignments (anaconda.com/download)
  - Related books, for example:
    - » Text Analytics with Python: A Practitioner's Guide to Natural Language Processing Paperback, 2019, by Dipanjan Sarkar
    - » Text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from Your Data, 2016, by Dipanjan Sarkar
    - » Applied Text Analysis with Python, Enabling Language-Aware Data Products with Machine Learning, by Benjamin Bengfort; Rebecca Bilbro; Tony Ojeda, Publisher: O'Reilly Media
    - » Python for Data Analysis, 3E by Wes McKinney (pub. yr. 2022) Free online access: https://wesmckinney.com/book/

Grading

- In-Class Exercises (best 10 of 12) — 10%
  - Individual, answered in a Quercus quiz that is open only during class
- Group Assignment 1 — SEC 10-K / 10-Q Parsing — 12%
  - Due Friday, October 23
- Group Assignment 2 — Sentiment Analysis (Keras ANN) — 16%
  - Due Friday, November 20
- Group Assignment 3 — Clustering & Topic Modelling — 12%
  - Due Friday, December 4
- Final Examination (in person, 3 hrs) — 50%
  - Open book: lecture materials and your own code. No generative AI, no internet.

- TA: Yufeng Zheng — yufeng.zheng@rotman.utoronto.ca

In-Class Exercises and Generative AI

- In-Class Exercises
  - Short individual questions during each class — some conceptual, some about a code output, chart, or diagram we generate live
  - Recorded in a Quercus quiz that opens only during the session; they cannot be completed remotely
  - Partial marks for a genuine attempt, full marks for a correct answer
  - Best 10 of 12 count, so an occasional missed class is already accommodated — no special consideration is granted

- Generative AI
  - Permitted on the group assignments, provided you disclose how you used it
  - Not permitted for the in-class exercises or the final examination
  - You are responsible for the correctness of everything you submit

Goals for the Course

- "*Learn how to process, classify, cluster, summarize, understand syntax, semantics and sentiment of text data with the power of Python!*"

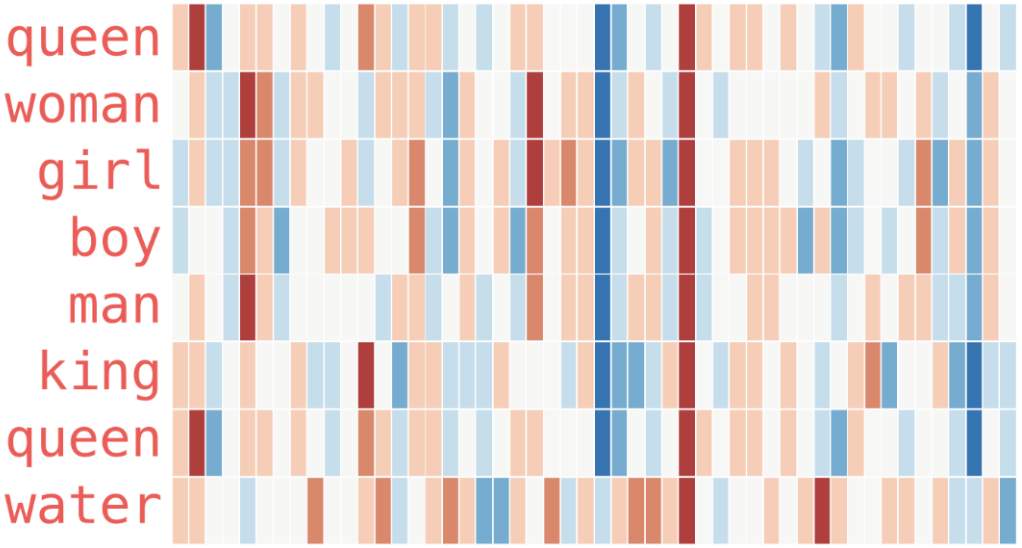
Dipanjan Sarkar

Text Data

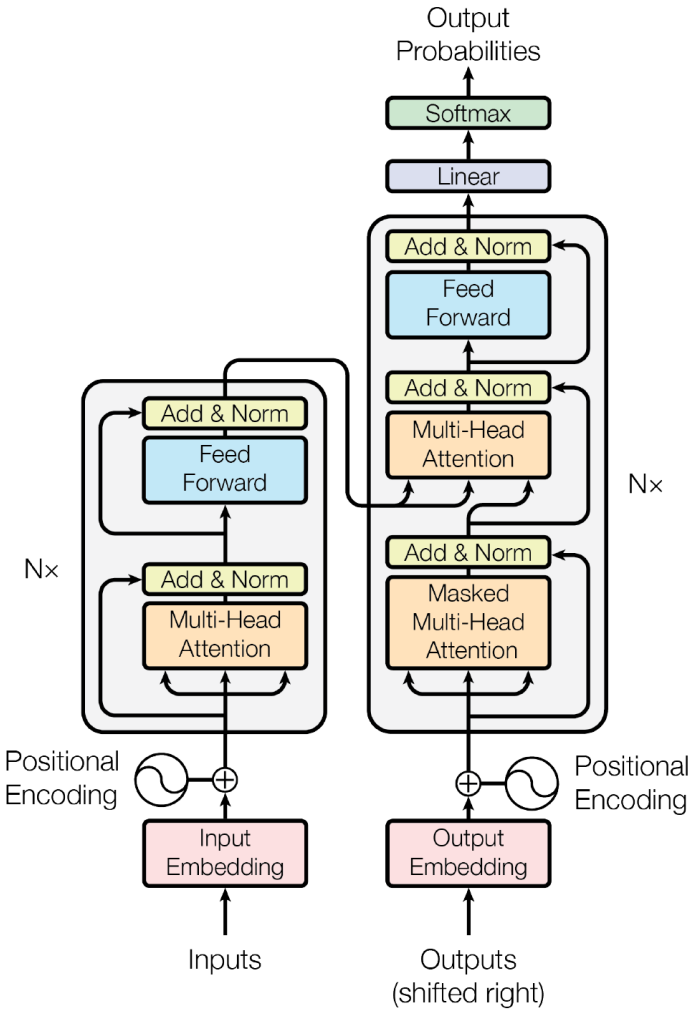
- In many other courses, e.g., RSM371, we/you have been dealing with **structured quantitative** data
  - Numerical
  - Binary (yes/no)
  - Multicategory – transformed into ordinal or usually better dummy/indicator variables
- In RSM317, we are working with **unstructured text**:
- **All** machine learning or deep learning models are "limited" because they **cannot** understand text data directly and they **only** understand **numeric** representations of features as inputs.

Even LLMs like ChatGPT **only** work with Numbers! Do you believe it?

Word2Vec Embeddings

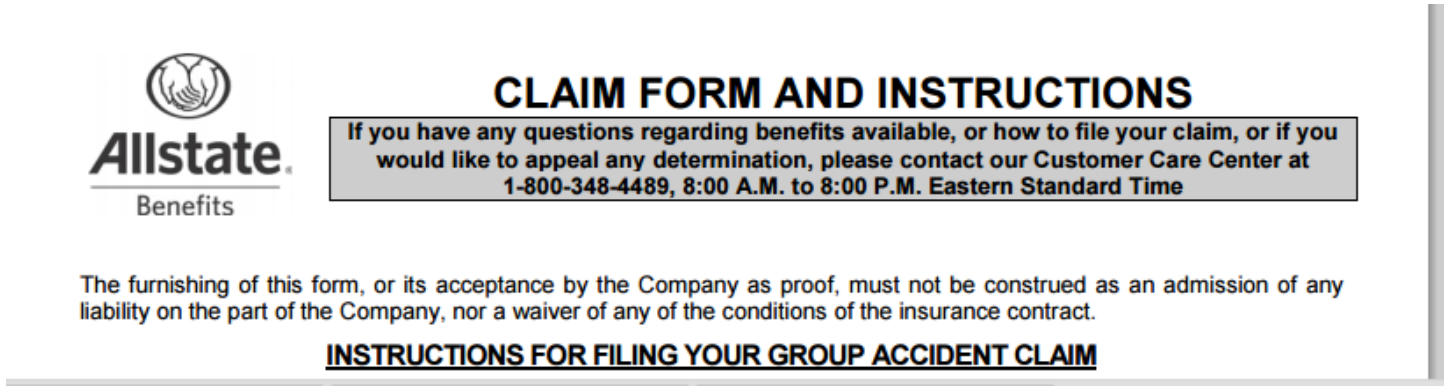


https://jalammar.github.io/illustrated-word2vec/



Applications of Text Mining

- Insurance fraud – notes in claim forms can be mined and transformed into predictor variables for a predictive model
- The model is trained on prior claims in two classes – found to be fraudulent, and not found to be fraudulent
- The model is then applied to new claims



Allstate Benefits

CLAIM FORM AND INSTRUCTIONS

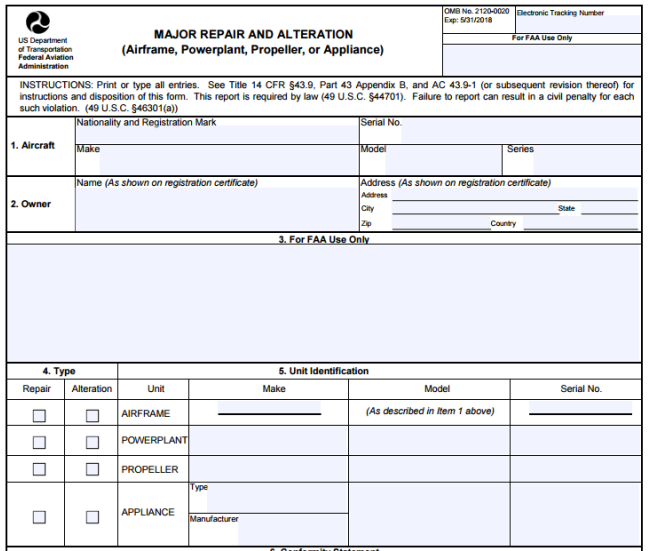
If you have any questions regarding benefits available, or how to file your claim, or if you would like to appeal any determination, please contact our Customer Care Center at 1-800-348-4489, 8:00 A.M. to 8:00 P.M. Eastern Standard Time

The furnishing of this form, or its acceptance by the Company as proof, must not be construed as an admission of any liability on the part of the Company, nor a waiver of any of the conditions of the insurance contract.

INSTRUCTIONS FOR FILING YOUR GROUP ACCIDENT CLAIM

Applications of Text Mining (cont'd)

- Maintenance or support tickets often contain text fields
- These fields could be mined to classify ticket in several ways:
  - What category of technician is needed to fix?
  - How much time to fix?
  - How urgent?



US Department of Transportation Federal Aviation Administration

MAJOR REPAIR AND ALTERATION
(Airframe, Powerplant, Propeller, or Appliance)

OMB No. 2120-0020 Exp: 5/31/2018 | Electronic Tracking Number | For FAA Use Only

INSTRUCTIONS: Print or type all entries. See Title 14 CFR §43.9, Part 43 Appendix B, and AC 43.9-1 (or subsequent revision thereof) for instructions and disposition of this form. This report is required by law (49 U.S.C. §44701). Failure to report can result in a civil penalty for each such violation. (49 U.S.C. §46301(a))

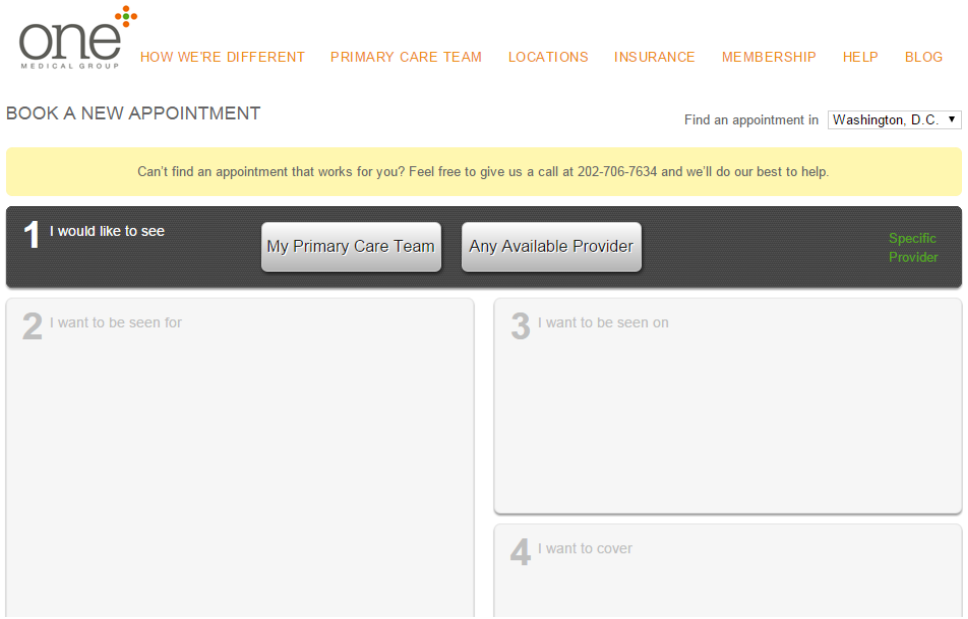
1. Aircraft	Nationality and Registration Mark	Serial No.	
	Make	Model	Series
2. Owner	Name *(As shown on registration certificate)*	Address *(As shown on registration certificate)* Address / City / State / Zip / Country	

3. For FAA Use Only

4. Type		**5. Unit Identification**			
Repair	Alteration	Unit	Make	Model	Serial No.
☐	☐	AIRFRAME		*(As described in Item 1 above)*	
☐	☐	POWERPLANT			
☐	☐	PROPELLER			
☐	☐	APPLIANCE	Type / Manufacturer		

Applications of Text Mining (cont'd)

- Medical triage/diagnosis
- Clinics could use patient online appointment request forms to route requests
  - Admin asst.
  - Nurse
  - Doctor



Applications of Text Mining
MMA Program at Rotman

- About 1/5 of the Practicum Projects use TM/NLP
- Deloitte:
- Examples:
  - Querying large data tables or databases using natural language (e.g., answering "how many fruits were bought last month?" based on a table with grocery receipt details).
  - Extended scope: answering questions containing implicit information (e.g., answering "how many green fruits were bought last month?" when the table has no information of fruit color, only fruit names).

Applications of Text Mining
MMA Program at Rotman (cont'd)

- AGF Investments LLC:
  - Financial disclosures – such as annual/quarterly reports – provide **comprehensive self-evaluation** about companies' recent performance. These timely reports provide **indispensable information** to value stocks and bonds. Even though investors have quick access to financial ratios, **tabulated information** are just the **tip of the iceberg**. **Narratives** from financial disclosures provide much more pivotal information. Deep dive into the context is necessary!
  - However, the size of digitalized financial reports is too tremendous for human to comprehend.

Applications of Text Mining
MMA Program at Rotman (cont'd)

- Infosys Consulting:
  - The consumer or commercial account opening process involves extracting information from various documents either provided by the customer or obtained from third-party databases.
  - Financial Institutions invest a lot of their resources in compiling the data and processing it to make an informed decision and this can be a highly manual process prone to errors and risks.
  - This project seeks to establish an AI-based process for validating the documents and extracting high-quality data and information from those documents.

Applications of Text Mining
MMA Program at Rotman (cont'd)

- Staples Canada ULC:
  - Enhancement of recommendation engine systems
  - Development of a Feature Mart

- Manulife:
  - Build a solution with GenAI to search HR Knowledge Articles and summarize retrieved information into answers that are easy to understand.

What really is Text Mining vs Natural Language Processing?

- For example, classify (or label) thousands of documents
  - Extension of predictive modeling
- Extract meaning from a single document – interpreting it like a human reads language
  - Natural Language Processing (NLP) (very ambitious)
    - automatic manipulation of natural language, like speech and text
    - around for more than 50 years (linguistics / computer linguistics)
    - problem of understanding text is **not** solved, and may never be
      - language is “messy”; few rules, many exceptions
      - “Alfred Hitchcock shot The Birds in Bodega Bay.”
  - While LLMs have revolutionized NLP, fundamental challenges like ambiguity, real-world reasoning, and linguistic messiness remain mostly unsolved. These models are tools, not true language "understanders“. Human oversight is critical for accuracy in complex applications.

Using Python and “Traditional” Tools for Text Analytics

Why are we not just using Large Language Models (LLMs) for all our text analysis tasks?

Because I want you to take
RSM416 - Generative AI for Data-Driven Management in Winter 2027 with me!



The Allure of Advanced Tools: The Sledgehammer for Nuts Effect

Advanced NLP Tools:

Large Language Models (LLMs) and Generative AI (e.g., GPT-4/5) are incredibly versatile and powerful.

Capable of tasks ranging from text generation to complex reasoning across domains.

The Drawback:

They are often viewed as the default choice for all NLP tasks, leading to inefficient use of resources.

Analogy:

It's like using a sledgehammer to crack a nut - impressive, but unnecessary for the task at hand.

Choosing the Right Tool for the Job

Traditional Tools:

TF-IDF for keyword extraction.

Logistic Regression for binary classification.

Clustering algorithms for unsupervised grouping.

When They Work Best:

Tasks where the problem is well-defined and does not require contextual understanding.

When Complexity Becomes Overengineering

Efficiency:

Simpler models require fewer computational resources.

Lower carbon footprint compared to training or querying LLMs.

Interpretability:

Easier to explain why a decision was made with models like Decision Trees or Naive Bayes.

Environmental Impact:

One GPT-3 training run
= driving a car for 1.2 million miles = 1.93 million km.

The Real Costs of Using a Sledgehammer for Nuts

Task	Traditional Models	LLMs
Training Costs	**<$100 on a single machine**	**$4M+ for GPT-3**
Inference Cost	**$0.001/query**	**$0.03–$0.06/query**
Compute Hardware	**Commodity CPU**	**GPUs/TPUs required**
CO_2 Emissions (Training)	**Negligible**	**550 metric tons**
Power Usage (Training)	**Standard electricity**	**1,287 MWh**

→ RSM317 → RSM416

Examples of Overkill

Spam Detection:

Efficient Solution: TF-IDF + Logistic Regression.

Overkill: LLMs.

Sentiment Analysis:

Efficient Solution: Naive Bayes with Bag of Words.

Overkill: LLMs.

Brief code example:

```
from sklearn.naive_bayes import MultinomialNB
nb = MultinomialNB()
nb.fit(X_train, y_train)
print("Accuracy:", nb.score(X_test, y_test))
```

When Advanced Tools Are Justified: Balancing Complexity and Efficiency

Context-Rich Tasks:

Answering nuanced questions or generating creative content.

Multilingual Analysis:

Handling data across multiple languages simultaneously.

Pipeline Complexity:

Use cases requiring multiple NLP capabilities (e.g., summarization, classification, translation).

Overengineering vs. Efficiency: A Cost-Benefit Perspective

Use Case	Traditional Tools	Advanced Tools
Keyword Extraction	TF-IDF, Bag of Words	LLMs (unnecessary)
Sentiment Analysis	Naive Bayes, Logistic Regression	LLMs (if nuanced sentiment required)
Fraud Detection in Insurance	Decision Trees, Random Forest	LLMs (if handling multilingual data)
Customer Support Summarization	LDA (Topic Modeling)	LLMs (more robust for summarizing diverse contexts)
Creative Writing or Content	Not feasible	LLMs
Multilingual Tasks	Limited options	LLMs (ideal for cross-lingual scenarios)

Avoid the Overkill Trap: Smarter, Not Just Bigger

Learn the Basics First:

Start with foundational tools to build understanding and efficiency.

Evaluate Costs and Benefits:

Consider time, resources, and environmental impact.

Avoid Overengineering:

Choose tools that match the task requirements without unnecessary complexity.

Using Python for Text Analytics

There are several *recommended* options to work on the coding examples:

1. On the lab's computer
   or your **own computer**: Install ANACONDA
   https://www.anaconda.com/download/ (Python 3.x)

   It comes with Jupyter Notebook and Spyder.

   -> Launch Jupyter Notebook

2. Use Google Colaboratory (runs in the **cloud**)
   https://colab.research.google.com
   Google's Jupyter Notebook
   You **need** a Google account!

Using Python for Text Analytics (cont'd)

There are several *recommended* options to work on the coding examples (cont'd):

3. UofT JupyterHub
   https://jupyter.utoronto.ca/
   – ONLY 2GB memory! ☹

https://highmem.datatools.utoronto.ca/hub/oauth_login
   – 5GB memory! ☺

4. DataLab by DataCamp
   https://www.datacamp.com/datalab

Using Python for Text Analytics (cont'd)

Let's code!

From Business Insider, September 2025:

Doug Steinberg, Software Engineer

"When it comes to my career, I think the only reason I'm any good is because I had to go through all that pain to learn how things actually work. If I didn't have to do that, I don't know if I would have had the discipline to actually learn it. If you don't really **know how things work**, you're not going to get that far. It's going to be a house of cards that will eventually fall over."

Using Python for Text Analytics (cont'd)

Bill Gates, August 2025:

Programming will continue to be a 100% human profession, even a century from now.
"Sure, AI can automate repetitive tasks like debugging code. But the heart of programming – **innovative problem-solving** and the crafting of **novel solutions** – depends on deep human insight. Writing code isn't just about typing commands; it's about understanding complex issues and devising tailored answers.

"Spam will be a thing of the past in two years' time."
Bill Gates, BBC News (24 January 2004)

Additional Readings

- Text Analytics with Python: A Practitioner's Guide to Natural Language Processing, Paperback, 2019, by Dipanjan Sarkar

Sources:

- Lambda Labs: GPT-3 Costs
  https://lambdalabs.com/blog/demystifying-gpt-3/
- MIT Tech Review: Carbon Cost of AI
  https://www.technologyreview.com/2019/06/06/239031/training-a-single-ai-model-can-emit-as-much-carbon-as-five-cars-in-their-lifetimes
- Nature: AI Water Usage
  https://www.nature.com/articles/s41893-023-01065-x
  https://arxiv.org/abs/2211.02001
- OpenAI Pricing
  https://openai.com/pricing
- AI Energy Consumption
  https://arxiv.org/abs/1906.02243